

Vitalijs Borisovs

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Computational Biologist & Imaging Specialist with a Ph.D. specializing in SBF-SEM, fluorescence microscopy, 3D reconstruction, and computational image analysis. Experienced with Arivis Pro, Zeiss ZEN, Python, and deep learning for biological imaging, automated segmentation, and quantitative analysis. Background in neurobiology and molecular biology, with experience in method development and research training.

Education

University of Latvia (2018 – 2023)

Ph.D., Biology

Thesis: *Genetic background of DNA damage in multiple sclerosis and free radical production in multiple sclerosis and type 1 diabetes*

Coursework in Molecular Biology, Pharmaceutical Bioinformatics, and Advanced Pharmaceutical Bioinformatics

Uppsala University (2014 – 2016)

M.Sc., Biochemistry

Thesis: *Determination of the enzymatic reaction rate-limiting step of alcohol dehydrogenase FucO*

Coursework in Drug Design, Protein chemistry, Biophysical chemistry, and Neuroscience

University of Latvia (2011 – 2014)

B.Sc., Chemistry

Thesis in Biochemistry and drug-target interactions.

Coursework in Physical Chemistry, Biochemistry, Analytical Chemistry, and Organic Chemistry

Teaching/supervising experience

Teaching Assistant | University of Latvia | 2018–2024

- Supervised undergraduate dissertation projects, providing guidance on research design, data analysis, and scientific writing.
- Delivered teaching sessions for B.Sc., M.Sc., and M.D. students in Biochemistry, Advanced Biochemistry, Genetics, and Dental Biomaterials.
- Contributed to course and programme development and student assessment.
- Assessed student coursework, examinations, and practical assignments.

Guest Lecturer | University of Latvia | 2020, 2023

- Delivered graduate-level lectures on Next-Generation Sequencing and Modern Science Technologies to Ph.D. and M.D. students.

Publications

- **V. Borisovs**, M. Bossi, L. Matino, P. Marmioli, G. Cavaletti. “Comparative Evaluation of En Bloc Staining Protocols for Deep-Learning Segmentation of Dorsal Root Ganglia in SBF-SEM” *J. Imaging* 2026, 12(10), 473
- **V. Borisovs**, M. Bossi, G. Cavaletti. “Morphometric analysis of axonal ultrastructure: Coordinated scaling of organelles and myelin.” *Micron*, 2026, 202, ISSN 0968-4328.
- **V. Borisovs**, G. Cavaletti. A Generalized Deep Learning Pipeline for Stain-Invariant Ultrastructural Segmentation in Peripheral Nerves. *J Imaging*. 2026 Jun 10;12(6):257. doi: 10.3390/jimaging12060257. PMID: 42346920; PMCID: PMC13301085.
- **V. Borisovs**, M. Bossi, L. Matino, P. Marmioli, G. Cavaletti. “New Approaches Based on Serial-Block Face Electron Microscopy to Investigate the Peripheral Nervous System.” *JPNS*, 2025, 30(2), e70019
- Rostoka E., Salna I., Dekante A., Pahirko L., **Borisovs V.**, Celma L., Valeinis J., Sjakste N., Sokolovska J. “DNA damage in leukocytes and serum nitrite concentration are negatively associated in type 1 diabetes.” *Mutagenesis*, 2021, 36(3):213-222
- **Borisovs V.**, Bodrenko J., Kalnina J., Sjakste N. “Nitrosative stress parameters and the level of oxidized DNA bases in patients with multiple sclerosis.” *Metabolic Brain Diseases*, 2021, 37(7):1935-1941.
- Leonova E., Shvirsksts K., **Borisovs V.**, Smelovs E., Sokolovska J., Bisenieks E., Duburs G., Grube M., Sjakste N. “Spectroscopic and electrochemical study of interactions between DNA and different salts of 1,4 dihydropyridine AV-153.” *PeerJ*, 2020, 8: e10061.

Conferences and presentations

- **V. Borisovs**, M. Bossi, G. Cavaletti. 3D Ultrastructural Analysis of Endoplasmic Reticulum-Mitochondria Interactions and Morphometric Scaling in Neuronal Axons, 78° Congresso Nazionale SIAI, Ravenna, Italy. 2025. Oral presentation (in Italian).
- **V. Borisovs**, J. Bodrenko, E. Leonova, I. Trapina, N. Paramonova, N. Sjakste. The MS puzzle: piecing together biochemical and genetic biomarkers, 7th Venusberg Meeting on Neuroinflammation, Belval, Luxembourg. 2023. Poster presentation. Abstract book.
- **V. Borisovs**, J. Bodrenko, E. Leonova, J. Kalniņa, N. Paramonova, N. Sjakste. Potential biomarkers for early diagnosis of multiple sclerosis, The 78th Conference of the University of Latvia, Riga, Latvia. 2020. Poster presentation. Abstract book.

Peer Review and Project Evaluation

- **Grant Project Reviewer, FISM / AISM (Italian Multiple Sclerosis Foundation) | 2026**
 - Appointed to evaluate the scientific merit, methodology, and feasibility of research proposals for the annual funding calls.
- **Peer Reviewer for scientific journals, including:**
 - *Metabolic Brain Diseases* (2021)
 - *PeerJ* (2021)
 - *Molecular and Cellular Biochemistry* (2023)
 - *Molecular Neurobiology* (2023)

Certifications

- ERC and MSC funding application writing, 2025, University of Milan – Bicocca.
- Machine Learning and Deep Learning engineering, 2025, Stanford University

Professional and Research experience

Research Associate, University of Milan – Bicocca (06.03.2024 – present)

- Perform 3D reconstruction and analysis of complex neurological structures, including peripheral nerves and Dorsal Root Ganglia (DRGs), using Serial Block-Face Scanning Electron Microscopy (SBF-SEM).
- Process and visualize large SBF-SEM datasets using Arivis Pro and Zeiss ZEN.
- Develop Python scripts for image preprocessing and analysis, including Contrast Limited Adaptive Histogram Equalization (CLAHE) to improve image quality and consistency for 3D reconstruction.
- Develop and train local deep learning models for automated segmentation and a custom statistical analysis pipeline for quantitative morphometric analysis of neuronal structures.
- Maintain a public GitHub repository with image-processing tools and deep learning models for nerve and DRG reconstruction.

Visiting Scientist | Luxembourg Centre for Systems Biomedicine | Sep 2023 – Mar 2024

- Isolated and cultured primary mouse microglia for in vitro experiments.
 - Performed phagocytosis assays using fluorescence and live-cell imaging.
 - Investigated NLRP3 inflammasome activation and amyloid- β nitration using biochemical and imaging-based assays.
 - Performed cytokine secretion assays using ELISA.
- Key skills: Primary cell culture, fluorescence imaging, live-cell imaging, cellular assays

Researcher | University of Latvia, Faculty of Medicine | 2016 – 2024

- Conducted research in molecular biology, genetics, and biochemical analysis.
- Progressed to Lead Researcher in 2023, coordinating research activities and contributing to project management.
- Performed DNA isolation, genotyping, and sequencing, as well as blood biomarker analysis.
- Supervised and taught undergraduate and graduate students.
- Contributed to research project planning, data analysis, and scientific reporting.

Field Application Scientist, R&D | MGI Tech | 2020 – 2022

- Managed the sample preparation team from 2021, coordinating laboratory activities and supporting method development.
- Developed and implemented DNA sequencing workflows and sample preparation solutions.
- Developed Python scripts for automated liquid-handling systems, improving laboratory workflow automation.
- Provided technical troubleshooting and support for sequencing workflows and instrumentation.
- Worked with customers and internal teams to evaluate requirements and develop technical solutions.
- Collaborated with Quality Assurance on SOPs, protocols, and laboratory guidelines.
- Trained and supported junior laboratory staff.

Scientific Assistant | Institute of Organic Synthesis | 2018 – 2020

- Studied protein–ligand interactions using NMR spectroscopy.
- Contributed to biochemical characterization and analysis of protein–ligand binding.

Research Intern (PHC OSMOSE Project) | Gustave Roussy Institute – Villejuif, France | 2018–2019

- Participated in an international research project investigating novel PARP1 and PARP2 inhibitors.
- Characterized 1,4-dihydropyridine (1,4-DHP) derivatives, with a focus on their DNA-binding properties and biochemical interactions.
- Completed two research placements as part of the project, one month in 2018 and one month in 2019.

Lead chemist | JSC Olainfarm | 2016 – 2017

- Quality control of drug substances
- Quality control method development for HPLC
- Quality control method validation

Quality Control Laboratory Assistant (Summer Internship) | JSC Grindeks | 2013

- Supported the method validation group within the quality control laboratory.
- Assisted with laboratory testing and routine analytical activities related to pharmaceutical quality control and method validation.

Core Technical & Research Expertise

- Imaging & 3D Reconstruction: Arivis Pro, Zeiss ZEN, SBF-SEM, confocal microscopy (Leica SP8, Zeiss Axio Imager 2), live-cell imaging, 3D image analysis and morphometric analysis.
- Computational Analysis: Python (custom scripts and data pipelines), machine learning and deep learning for biological image segmentation, statistical analysis, automated morphometric analysis, Git/GitHub.
- Molecular & Cellular Biology: NLRP3 inflammasome assays, primary microglia isolation and phagocytosis assays (mouse models), mammalian and iPSC cell culture, next-generation sequencing (Illumina/BGI; scRNA-seq, RNA-seq, WGS/WES), proteomics (IMAC, NMR docking), and biomarker assays (ELISA, ROS).

Languages

- English – proficient
- Latvian – proficient
- German – elementary
- Swedish – intermediate
- Italian - intermediate
- Russian – advanced